

Symbolic Projection Notes for $B_{c1} \rightarrow B_c^{(*)}\gamma$

Working notes

July 12, 2026

1 Scope

These notes collect the compact FeynCalc projection results for the heavy-heavy channels

$$B_{c1}(6743) \rightarrow B_c\gamma, \quad B_{c1}(6743) \rightarrow B_c^*\gamma, \quad B_{c1}(6750) \rightarrow B_c\gamma, \quad B_{c1}(6750) \rightarrow B_c^*\gamma.$$

The calculation is structurally different from the D_s and B_s light-cone sum rules. In the B_c system both valence constituents are heavy, so the leading radiative OPE is a hard-photon heavy-heavy calculation. There is no light spectator line to be replaced by photon distribution amplitudes, and no $\chi_s\langle\bar{s}s\rangle$ type term.

2 Currents and Propagators

For $B_c^+ \sim c\bar{b}$, we use

$$j_5 = \bar{b}i\gamma_5c, \quad j_\rho^V = \bar{b}\gamma_\rho c,$$

and the two axial basis currents

$$J_\mu^A = \bar{b}\gamma_\mu\gamma_5c, \quad J_\mu^B = \frac{i}{m_b + m_c}\bar{b}\sigma_{\mu\alpha}P^\alpha\gamma_5c, \quad P = p + q.$$

The physical currents are

$$J_{1\mu} = \sin\theta J_\mu^A + \cos\theta J_\mu^B, \quad J_{2\mu} = \cos\theta J_\mu^A - \sin\theta J_\mu^B.$$

This mixing is used at the invariant-amplitude level. The OPE is first derived for the basis currents because the two Dirac traces are independent; the physical amplitudes are then

$$\hat{\Pi}_1 = \sin\theta \hat{\Pi}_A + \cos\theta \hat{\Pi}_B, \quad \hat{\Pi}_2 = \cos\theta \hat{\Pi}_A - \sin\theta \hat{\Pi}_B.$$

For the pseudoscalar final state,

$$g_i = \frac{m_b + m_c}{m_{B_c}^2 f_{B_c} m_i f_i} \exp\left(\frac{m_i^2}{M_1^2} + \frac{m_{B_c}^2}{M_2^2}\right) \hat{\Pi}_i^{(P)},$$

while for the vector final state,

$$g_i^* = \frac{1}{m_{B_c^*} f_{B_c^*} m_i f_i} \exp\left(\frac{m_i^2}{M_1^2} + \frac{m_{B_c^*}^2}{M_2^2}\right) \hat{\Pi}_i^{(V)}.$$

The updated scripts apply this rotation with physical two-point residues f_1 and f_2 . The vector $B_c^*\gamma$ entries use the standard vector-current normalization, with the alternative projector normalization kept only as a convention check.

The free heavy-quark propagator in momentum space is

$$S_Q^{(0)}(k) = \frac{i(\not{k} + m_Q)}{k^2 - m_Q^2 + i0}, \quad Q = c, b.$$

The hard photon is included by inserting the electromagnetic vertex on either heavy line:

$$S_Q^{(\gamma)}(k, k+q) = S_Q^{(0)}(k+q) ie_Q \gamma_\nu \epsilon^\nu S_Q^{(0)}(k).$$

For the $\bar{b}$ line it is convenient to use the effective antiquark charge $e_{\bar{b}} = +e/3$, while $e_c = +2e/3$. Equivalently, one may keep $e_b = -e/3$ and track the orientation of the antiquark line. The scripts use $e_{\bar{b}}$ explicitly.

Thus, yes: this is the same heavy-heavy propagator logic as in a B_c -mixing sum-rule calculation. The important difference from the strange-sector radiative calculation is that we do not insert a light-quark photon DA. If we later include power corrections, the natural heavy-heavy correction is the background-gluon expansion of the heavy propagators, e.g. gluon-condensate terms, not strange-quark condensate photon DAs.

The submitted B_c -mixing work uses the same local basis,

$$J_\mu^A = \bar{b} \gamma_\mu \gamma_5 c, \quad J_\mu^B = \frac{i}{m_b + m_c} \bar{b} \sigma_{\mu\alpha} p^\alpha \gamma_5 c,$$

with $m_b = 4.18$ GeV, $m_c = 1.27$ GeV, and the final working window

$$7 \text{ GeV}^2 \leq M^2 \leq 9 \text{ GeV}^2, \quad 53 \text{ GeV}^2 \leq s_0 \leq 55 \text{ GeV}^2.$$

Those are useful starting values for the heavy-heavy OPE in the radiative problem. They are not yet automatically the final radiative-transition windows; the final windows must be fixed by the three-point sum-rule stability, pole dominance, and convergence tests.

3 Step-by-Step Sum-Rule Construction

This section is written as a checklist for reproducing the calculation.

1. Start from basis currents. The physical currents J_1, J_2 are mixed combinations of J_A, J_B . We derive the OPE for J_A and J_B separately only because the Dirac traces are simpler. The physical amplitudes are obtained afterward through the $\sin \theta, \cos \theta$ rotation.

2. Write the correlators. For $B_c \gamma$,

$$\Pi_\mu^{(5,K)}(p, q) = i \int d^4x e^{ipx} \langle \gamma(q, \epsilon) | T \{ \bar{b}(x) i \gamma_5 c(x) J_{K\mu}^\dagger(0) \} | 0 \rangle, \quad K = A, B.$$

For $B_c^* \gamma$,

$$\Pi_{\mu\nu}^{(V,K)}(p, q) = i \int d^4x e^{ipx} \langle \gamma(q, \epsilon) | T \{ \bar{b}(x) \gamma_\nu c(x) J_{K\mu}^\dagger(0) \} | 0 \rangle.$$

3. Contract the heavy fields. There are two hard-photon diagrams: photon emission from the c line and photon emission from the $\bar{b}$ line. The free heavy propagator is

$$S_Q^{(0)}(k) = \frac{i(\not{k} + m_Q)}{k^2 - m_Q^2 + i0},$$

and the photon insertion is

$$S_Q^{(\gamma)}(k, k+q) = S_Q^{(0)}(k+q) ie_Q \not{\epsilon} S_Q^{(0)}(k).$$

The charge convention in the scripts is $e_c = +2/3$ and $e_{\bar{b}} = +1/3$. No photon DA or light-quark condensate term appears in this heavy-heavy leading baseline.

4. Evaluate and project the traces. For $B_c\gamma$ project onto

$$S_{\mu\nu} = p_\nu q_\mu - (p \cdot q)g_{\mu\nu}, \quad \mathcal{P}^{\mu\nu} = \frac{S^{\mu\nu}}{2(p \cdot q)^2}, \quad S_{\mu\nu}\mathcal{P}^{\mu\nu} = 1.$$

For $B_c^*\gamma$ the current diagnostic tensor is

$$V_{\mu\nu\rho} = p_\nu \epsilon_{\mu\rho pq} - (p \cdot q)\epsilon_{\mu\nu\rho p}, \quad V_{\mu\nu\rho}V^{\mu\nu\rho} = -4p^2(p \cdot q)^2.$$

The vector-channel result still needs matching to the physical $1^+ \rightarrow 1^- \gamma$ amplitude basis.

5. Reduce to denominator structures. For photon emission from the charm line use

$$D_1 = (k + q)^2 - m_c^2, \quad D_2 = k^2 - m_c^2, \quad D_3 = (k - p)^2 - m_b^2.$$

For photon emission from the anti-bottom line use

$$D_1 = k^2 - m_c^2, \quad D_2 = (k - p)^2 - m_b^2, \quad D_3 = (k - p + q)^2 - m_b^2.$$

The triangle pieces contain the full three-denominator product. Terms where a numerator cancels one denominator become two-denominator/contact terms and must be included or shown to vanish.

6. Borel match and rotate. The true variables are p^2 and $P^2 = (p + q)^2$, with

$$p \cdot q = \frac{P^2 - p^2}{2}.$$

The pilot numerical calculation uses the diagonal choice $M_1^2 = M_2^2 = 2M^2$, leading to the practical exponential

$$\exp\left[\frac{m_i^2 + m_f^2}{2M^2}\right].$$

After obtaining $\hat{\Pi}_A, \hat{\Pi}_B$, the physical combinations are

$$\hat{\Pi}_{6743} = \sin\theta \hat{\Pi}_A + \cos\theta \hat{\Pi}_B, \quad \hat{\Pi}_{6750} = \cos\theta \hat{\Pi}_A - \sin\theta \hat{\Pi}_B.$$

For the $B_c\gamma$ widths we use

$$\Gamma(1^+ \rightarrow 0^- \gamma) = \frac{\alpha_{\text{em}}}{3} g^2 E_\gamma^3, \quad E_\gamma = \frac{m_i^2 - m_f^2}{2m_i}.$$

4 $B_{c1} \rightarrow B_c\gamma$: E1 Projection

For the pseudoscalar final state we project onto

$$S_{\mu\nu} = p_\nu q_\mu - (p \cdot q)g_{\mu\nu}, \quad \mathcal{P}_{\mu\nu} = \frac{S_{\mu\nu}}{2(p \cdot q)^2}.$$

FeynCalc gives the normalization check

$$S_{\mu\nu}\mathcal{P}^{\mu\nu} = 1.$$

We denote

$$p^2 = p_2, \quad p \cdot q = pq.$$

After rewriting loop scalar products in terms of inverse denominators and setting those denominators to zero, the projected triangle-core numerators are

$$C_A^{(c)} = -4i m_c + \frac{4i m_b m_c^2}{pq} - \frac{4i m_c^3}{pq},$$

$$C_A^{(\bar{b})} = -4i m_b + \frac{4i m_b^3}{pq} - \frac{4i m_b^2 m_c}{pq},$$

so that

$$C_A = e_c C_A^{(c)} + e_{\bar{b}} C_A^{(\bar{b})}.$$

For the tensor current,

$$C_B^{(c)} = \frac{4i m_b m_c - 8i m_c^2}{m_b + m_c} - \frac{4i m_c^2 p_2}{(m_b + m_c)pq},$$

$$C_B^{(\bar{b})} = -\frac{4i m_b m_c}{m_b + m_c} - \frac{4i m_b^2 p_2}{(m_b + m_c)pq},$$

and

$$C_B = e_c C_B^{(c)} + e_{\bar{b}} C_B^{(\bar{b})}.$$

The denominator-cancellation terms are stored in the corresponding output file. They should not be discarded automatically; they correspond to reduced two-denominator or contact contributions and must be treated in the dispersion/Borel analysis.

5 $B_{c1} \rightarrow B_c^* \gamma$: Levi-Civita Projection

For the vector final state a useful gauge-invariant structure is

$$V_{\mu\nu\rho} = p_\nu \epsilon_{\mu\rho pq} - (p \cdot q) \epsilon_{\mu\nu\rho p}.$$

It is transverse in the photon index for $q^2 = 0$. The FeynCalc checks are

$$V_{\mu\nu\rho} V^{\mu\nu\rho} = -4p_2(pq)^2, \quad V_{\mu\nu\rho} \mathcal{P}^{\mu\nu\rho} = 1.$$

The projected J_A triangle cores are

$$C_{A,V}^{(c)} = -\frac{4i m_b m_c}{p_2} + \frac{2i m_c^2}{p_2} + \frac{4i m_c^2}{pq},$$

$$C_{A,V}^{(\bar{b})} = -\frac{2i m_b^2}{p_2} + \frac{4i m_b m_c}{p_2} + \frac{4i m_b^2}{pq}.$$

The tensor-current vector cores are

$$C_{B,V}^{(c)} = \frac{2i m_c}{m_b + m_c} + \frac{2i m_b^2 m_c - 4i m_b m_c^2 + 2i m_c^3}{(m_b + m_c)p_2}$$

$$- \frac{4i m_b m_c^2 - 4i m_c^3}{(m_b + m_c)pq} + \frac{2i m_c pq}{(m_b + m_c)p_2},$$

$$C_{B,V}^{(\bar{b})} = \frac{2i m_b}{m_b + m_c} + \frac{-6i m_b^3 + 4i m_b^2 m_c + 2i m_b m_c^2}{(m_b + m_c)p_2}$$

$$- \frac{4i m_b^3 - 4i m_b^2 m_c}{(m_b + m_c)pq} + \frac{2i m_b pq}{(m_b + m_c)p_2}.$$

Again,

$$C_{A,V} = e_c C_{A,V}^{(c)} + e_{\bar{b}} C_{A,V}^{(\bar{b})}, \quad C_{B,V} = e_c C_{B,V}^{(c)} + e_{\bar{b}} C_{B,V}^{(\bar{b})}.$$

6 Controlled $B_c\gamma$ Pass

The script `Bc_gamma/scripts/bc_ps_complete_analysis.py` uses the compact $B_c\gamma$ triangle cores above and replaces the earlier common axial-vector normalization by physical two-point residues. We use the same central heavy-quark inputs and Borel window as the submitted B_c -mixing analysis,

$$m_b = 4.18 \text{ GeV}, \quad m_c = 1.27 \text{ GeV}, \quad \theta = 43.3^\circ,$$

more precisely $\theta_{\text{mean}} = 43.295^\circ$ with $\sigma_\theta = 0.151^\circ$ and 16–84% interval $[43.147^\circ, 43.455^\circ]$ from the B_c -mixing Monte Carlo. The updated Monte Carlo samples this θ distribution.

$$M^2 = \{7, 8, 9\} \text{ GeV}^2, \quad s_0 = \{53, 54, 55\} \text{ GeV}^2.$$

The pseudoscalar decay-constant benchmark is

$$f_{B_c} = 0.371 \pm 0.037 \text{ GeV}.$$

The physical axial-vector residues are extracted from the same two-point B_c -mixing basis. With $s_\theta = \sin \theta$ and $c_\theta = \cos \theta$,

$$\Pi_1 = s_\theta^2 \Pi_{AA} + 2s_\theta c_\theta \Pi_{AB} + c_\theta^2 \Pi_{BB}, \quad \Pi_2 = c_\theta^2 \Pi_{AA} - 2s_\theta c_\theta \Pi_{AB} + s_\theta^2 \Pi_{BB},$$

and

$$f_i(M^2, s_0) = \left[\frac{e^{m_i^2/M^2} \Pi_i(M^2, s_0)}{m_i^2} \right]^{1/2}.$$

Using the perturbative heavy-heavy two-point spectral densities over the window above gives

$$f_1 = 0.264_{-0.005}^{+0.005} \text{ GeV}, \quad f_2 = 0.722_{-0.013}^{+0.016} \text{ GeV}$$

from the 16–84% grid interval. This replaces the older common benchmark $f_{B_{c1}} = 0.373 \text{ GeV}$.

Table 1: Updated $B_c\gamma$ results with physical two-point residues and the explicitly integrated all-positive dimension-four G^2 sector. The quoted intervals are the Monte Carlo 16–84% ranges.

Channel	Status	Coupling/projection coefficient	Γ (keV)
$B_{c1}(6743) \rightarrow B_c\gamma$	controlled $B_c\gamma$	$+0.0487 [0.0414, 0.0572] \text{ GeV}^{-1}$	$0.532 [0.385, 0.734]$
$B_{c1}(6750) \rightarrow B_c\gamma$	controlled $B_c\gamma$	$-0.206 [-0.230, -0.188] \text{ GeV}^{-1}$	$10.04 [8.29, 12.45]$

For the vector final state we also remove the fixed-normalization assumption. The final-state vector decay constant is defined by

$$\langle 0 | \bar{c} \gamma_\rho b | B_c^*(p, \eta) \rangle = m_{B_c^*} f_{B_c^*} \eta_\rho.$$

Projecting the two-point vector-current correlator onto the spin-1 invariant gives the perturbative density

$$\rho_V(s) = \frac{3}{16\pi^2} \frac{\sqrt{\lambda(s, m_b^2, m_c^2)}}{s} \left[-\frac{8(k \cdot p)^2}{s} - 4m_c^2 + 12(k \cdot p) + 12m_b m_c \right],$$

where $k \cdot p = (s + m_c^2 - m_b^2)/2$. In the equal-mass limit this reduces to the expected S -wave vector-current density proportional to $(s + 2m_Q^2) \sqrt{1 - 4m_Q^2/s}$. The corresponding two-point sum rule is

$$f_{B_c^*}^2 m_{B_c^*}^2 e^{-m_{B_c^*}^2/M_V^2} = \int_{(m_b+m_c)^2}^{s_0, V} ds e^{-s/M_V^2} \rho_V(s).$$

Using $M_V^2 = 5\text{--}8 \text{ GeV}^2$ and $s_{0,V} = 40\text{--}44 \text{ GeV}^2$, the direct spin-projector density gives

$$f_{B_c^*}^{\text{projector}} = 0.753 [0.693, 0.808] \text{ GeV}.$$

This is a projector-convention diagnostic. For the physical matrix element $\langle 0 | \bar{c} \gamma_\rho b | B_c^* \rangle = m_{B_c^*} f_{B_c^*} \eta_\rho$, the standard vector-invariant convention divides the density by the transverse spin-trace factor of three. The corresponding self-contained two-point result is

$$f_{B_c^*}^{\text{standard}} = 0.435 [0.400, 0.467] \text{ GeV}.$$

This is close to the commonly used reference value $f_{B_c^*} = 0.384 \pm 0.038 \text{ GeV}$. We therefore quote the standard normalization as the preferred self-contained result and keep the reference normalization as a sensitivity check:

Table 2: Hard-QCDSR $B_c^* \gamma$ predictions. The standard vector-invariant entries include the perturbative hard-photon three-point term, the physical f_1, f_2 residues, and the explicitly integrated all-positive dimension-four G^2 sector. The reference-normalization rows are retained as perturbative sensitivity checks.

Channel	Normalization	Coupling/projection coefficient	Γ (keV)
$B_{c1}(6743) \rightarrow B_c^* \gamma$	standard vector-invariant	$-0.0764 [-0.0821, -0.0717] \text{ GeV}^{-1}$	$70.3 [61.9, 81.2]$
$B_{c1}(6750) \rightarrow B_c^* \gamma$	$f_{B_c^*}$ standard vector-invariant	$+0.173 [0.161, 0.185] \text{ GeV}^{-1}$	$382 [331, 437]$
$B_{c1}(6743) \rightarrow B_c^* \gamma$	$f_{B_c^*}$ reference $f_{B_c^*} = 0.384 \pm 0.038 \text{ GeV}$	$-0.0877 [-0.0986, -0.0795] \text{ GeV}^{-1}$	$92.6 [76.1, 117]$
$B_{c1}(6750) \rightarrow B_c^* \gamma$	$f_{B_c^*}$ reference $f_{B_c^*} = 0.384 \pm 0.038 \text{ GeV}$	$+0.187 [0.169, 0.210] \text{ GeV}^{-1}$	$447 [365, 564]$

The $B_c \gamma$ entries are the controlled hard-photon baseline from the present calculation. The vector entries are no longer just placeholders: they use the $V_{\mu\nu\rho}$ projection, the physical f_1, f_2 residues, and an explicit $f_{B_c^*}$ normalization. The projector diagnostic gives $24.8[21.8, 28.5]$ and $120[104, 137] \text{ keV}$ for the two vector channels and is kept only in the output CSV files as a convention check. The table above uses the standard vector-current convention. After the all-positive G^2 integration below, these entries become the preferred final vector predictions; the projector diagnostic remains only a normalization check.

The vector stability plots are generated by `bc_vec_complete_analysis.py`. On the central $3 \times 3 (M^2, s_0)$ grid, the standard-normalization widths vary from about 66.7 to 80.9 keV for $B_{c1}(6743) \rightarrow B_c^* \gamma$, and from about 310 to 406 keV for $B_{c1}(6750) \rightarrow B_c^* \gamma$. The large higher state width is therefore not a numerical instability of the selected Borel window.

Before the explicit all-positive calculation was completed, we used the same conservative screening logic as in the $B_c \gamma$ channel, but with one additional residue factor. The vector sum rule contains $f_i f_{B_c^*}$ in the denominator, so

$$\left| \frac{\delta g}{g} \right| \lesssim \left| \frac{\delta \Pi^{(3)}}{\Pi^{(3)}} \right| + \left| \frac{\delta f_i}{f_i} \right| + \left| \frac{\delta f_{B_c^*}}{f_{B_c^*}} \right| \simeq 2 \max \left| \frac{\Pi_{G^2}}{\Pi_{\text{pert}}} \right|,$$

and

$$\left| \frac{\delta \Gamma}{\Gamma} \right| \lesssim 4 \max \left| \frac{\Pi_{G^2}}{\Pi_{\text{pert}}} \right|.$$

For the preferred standard vector normalization this gives the screening envelopes

channel	median envelope	maximum envelope
$B_{c1}(6743) \rightarrow B_c^* \gamma$	$\pm 11.1 \text{ keV}$	$\pm 27.7 \text{ keV}$
$B_{c1}(6750) \rightarrow B_c^* \gamma$	$\pm 53.6 \text{ keV}$	$\pm 139 \text{ keV}$

for the widths. This was not an explicit condensate correction; it is retained only as a diagnostic explaining why the all-positive G^2 sector was computed. The refined contact audit below shows that the contact rows do not have ordinary physical two-channel support; the crossed rows are retained only as an audit trail.

7 Comparison with Experiment and Other Theory

LHCb has observed a wide peaking structure in the $B_c^+\gamma$ mass spectrum with significance above seven standard deviations. A minimal two-peak description gives [1, 2]

$$M_1 = 6704.8 \pm 5.5 \pm 2.8 \pm 0.3 \text{ MeV}, \quad M_2 = 6752.4 \pm 9.5 \pm 3.1 \pm 0.3 \text{ MeV},$$

where the errors are statistical, systematic, and due to the B_c^+ mass. No individual radiative partial width is measured. The natural widths are expected to be only a few hundred keV and are neglected in the experimental fit because they are small compared with the photon-energy resolution.

Table 3: Qualitative comparison with the current experimental information from LHCb [1, 2].

Quantity	Experiment	Present input/result
Observed $B_c(1P)^+$ -like signal	yes, $> 7\sigma$ in $B_c^+\gamma$	channels studied explicitly
Lower visible peak	$6704.8 \pm 5.5 \pm 2.8 \pm 0.3 \text{ MeV}$	input state 6743 MeV
Higher visible peak	$6752.4 \pm 9.5 \pm 3.1 \pm 0.3 \text{ MeV}$	input state 6750 MeV
Partial widths	not measured	$B_c\gamma$: 0.532, 10.04 keV; $B_c^*\gamma$: 70.3, 382 keV

The lower visible peak should not be identified one-to-one with the input 6743 MeV state without care. In the measured $B_c\gamma$ spectrum, decays through $B_c^*\gamma$, followed by an unreconstructed soft $B_c^* \rightarrow B_c\gamma$, can shift visible peaks by the unknown hyperfine splitting.

For comparison with other theory we should not compress different papers into a single range. Table 4 lists each quoted calculation separately. The entries are the values collected in Bondar–Milstein [4]; the original model papers are cited in the first column.

Table 4: Comparison with representative predictions for $B_{c1} \rightarrow B_c^{(*)}\gamma$. Widths are in keV.

Publication	$6743 \rightarrow B_c\gamma$	$6743 \rightarrow B_c^*\gamma$	$6750 \rightarrow B_c\gamma$	$6750 \rightarrow B_c^*\gamma$
Godfrey [5]	13	60	80	11
Ebert–Faustov–Galkin [6]	18.4	78.9	132	13.6
Fulcher [7]	32.4	75.6	127.8	26.2
Gershtein et al. [8]	11.6	77.8	131.1	8.1
T. Y. Li et al. [9]	16.6	49	66.6	10.5
Q. Li et al. [10]	35	70	74	40
X. J. Li et al. [11]	30.1	47.8	64	25.6
Eichten–Quigg [12]	9.9	62.5	92.3	7.5
Bondar–Milstein [4]	22	75	47	2
This work	0.532[0.385, 0.734]	70.3[61.9, 81.2]	10.04[8.29, 12.45]	382[331, 437]

Thus the $B_c\gamma$ and $B_c^*\gamma$ channels are controlled hard-QCDSR predictions including the explicitly integrated all-positive G^2 sector. The contact-support audits leave no ordinary two-channel contact contribution to add to the standard physical double-Borel sum.

8 Perturbative and Condensate Content

The $B_c\gamma$ numerical table above contains the leading hard-photon perturbative three-point baseline, normalized with perturbative two-point physical residues. In practical terms this means:

- the photon is emitted from the charm or anti-bottom heavy-quark line;

- both heavy propagators are free propagators in the radiative three-point OPE;
- the retained spectral density is the perturbative triangle hard contribution generated by the projected Dirac traces;
- no photon DA, $\chi\langle\bar{q}q\rangle$, or light-quark condensate contribution appears, because there is no light valence quark in B_c ;
- gluon-condensate corrections from the background-gluon expansion of the heavy propagators are included through the explicit all-positive G^2 sector;
- reduced two-denominator/contact terms from denominator cancellation have been audited in the vector $B_c^*\gamma$ channel and do not give ordinary two-channel double-pole contributions.

Thus the present $B_c\gamma$ and $B_c^*\gamma$ results are best described as controlled hard-QCDSR predictions with the explicit all-positive G^2 sector included. The contact audits leave only crossed-contact or single-virtuality candidates outside the standard two-channel projection.

9 Dimension-4 Radiative Gluon-Condensate Workbench

The heavy-heavy B_c system has no photon DA or light-quark condensate contribution, but the background-gluon expansion of the heavy propagators generates possible dimension-4 gluon-condensate corrections to the radiative three-point function. To keep the calculation reproducible, the $\langle\alpha_s G^2/\pi\rangle$ part was split into small FeynCalc chunks instead of reducing one very large trace.

For the $B_c\gamma$ E1 structure the two hard-photon topologies are

$$S_c(k+q)\gamma_{\text{em}}S_c(k)\gamma_5S_b(k-p), \quad S_c(k)\gamma_5S_b(k-p)\gamma_{\text{em}}S_b(k-p+q).$$

In each topology we considered

- one single-line insertion $S_Q^{(G^2)}$ on each heavy propagator segment;
- one open-gluon-pair insertion $S^{(G)}S^{(G)}$ for each pair of propagator segments.

Thus there are six single-line topologies and six open-pair topologies. Each topology is evaluated for the two basis currents J_A and J_B , giving 24 projected numerator rows in total.

The projection uses

$$S_{\mu\nu} = p_\nu q_\mu - (p \cdot q)g_{\mu\nu}, \quad \mathcal{P}^{\mu\nu} = \frac{S^{\mu\nu}}{2(p \cdot q)^2},$$

so that the projected coefficient is the coefficient of the same E1 tensor used in the perturbative baseline. With the present conventions the audit is

class	current-split rows	projected terms
$S_Q^{(G^2)}$	12	221
$S^{(G)}S^{(G)}$	12	164
total	24	385

The scripts which generate these checks are

- Bc_gamma/scripts/step3_ps_g2_topology_inventory.py;
- Bc_gamma/scripts/step3_ps_g2_single_line_cphoton.wl;

- Bc_gamma/scripts/step3_ps_g2_single_line_bbarphoton.wl;
- Bc_gamma/scripts/step3_ps_g2_open_pair_cphoton.wl;
- Bc_gamma/scripts/step3_ps_g2_open_pair_bbarphoton.wl;
- Bc_gamma/scripts/step3_ps_g2_projection_summary.py;
- Bc_gamma/scripts/step3_ps_g2_denominator_bookkeeping.py.

The raw outputs are stored in Bc_gamma/outputs/step3_ps_g2_*.txt and the summary table is stored in Bc_gamma/outputs/step3_ps_g2_projection_summary.csv. The denominator manifest is stored in Bc_gamma/outputs/step3_ps_g2_denominator_bookkeeping.csv.

The denominator bookkeeping is most compact in Schwinger parameters. For the charm-line photon topology we use

$$a \leftrightarrow S_c(k+q), \quad b \leftrightarrow S_c(k), \quad r \leftrightarrow S_b(k-p), \quad A = a+b+r.$$

After shifting the loop momentum the quadratic form is

$$\Phi_c = \frac{r(a+b)}{A}p^2 + \frac{2ar}{A}(p \cdot q) - (a+b)m_c^2 - rm_b^2 = \frac{rb}{A}p^2 + \frac{ar}{A}P^2 - (a+b)m_c^2 - rm_b^2.$$

For the anti-bottom-line photon topology we use

$$a \leftrightarrow S_c(k), \quad b \leftrightarrow S_b(k-p), \quad r \leftrightarrow S_b(k-p+q), \quad A = a+b+r,$$

which gives

$$\Phi_{\bar{b}} = \frac{a(b+r)}{A}p^2 + \frac{2r(b+r)}{A}(p \cdot q) - am_c^2 - (b+r)m_b^2.$$

Equivalently, since $P^2 = p^2 + 2p \cdot q$,

$$\Phi_{\bar{b}} = \frac{(b+r)(a-r)}{A}p^2 + \frac{r(b+r)}{A}P^2 - am_c^2 - (b+r)m_b^2.$$

For $S_Q^{(G^2)}$, the selected propagator power is four and the other two propagator powers are one. For $S^{(G)}S^{(G)}$, the selected pair carries powers two and two, while the remaining propagator has power one.

This completes the numerator-projection stage of the radiative gluon-condensate workbench and attaches denominator powers to each projected row. It is not yet a numerical condensate correction to the decay width. The next step is to perform the double-dispersion/double-Borel reduction and integrate the result over the same Borel and continuum-threshold windows used for the perturbative baseline.

Conservative G^2 Size Screen

Before completing the full derivative-delta reduction, we performed a conservative size screen using the completed B_c -mixing OPE convergence table in the same numerical window,

$$M^2 = \{7, 8, 9\} \text{ GeV}^2, \quad s_0 = \{53, 54, 55\} \text{ GeV}^2.$$

The script is bc_ps_g2_screening_estimate.py. The corresponding CSV and text outputs are stored in the Bc_gamma/outputs directory.

The screen defines

$$\Delta_{2\text{pt}}^{G^2} = \max \left(\left| \frac{\Pi_{AA}^{G^2}}{\Pi_{AA}^{\text{pert}}} \right|, \left| \frac{\Pi_{AB}^{G^2}}{\Pi_{AB}^{\text{pert}}} \right|, \left| \frac{\Pi_{BB}^{G^2}}{\Pi_{BB}^{\text{pert}}} \right| \right)$$

from the B_c -mixing calculation. Since $f \sim \sqrt{\Pi}$, the normalization shift is estimated as $\Delta_f \simeq \Delta_{2\text{pt}}^{G^2}/2$. Taking the unknown radiative three-point G^2 shift to be of the same size gives the conservative envelope

$$\left| \frac{\delta g}{g} \right|_{\text{env}} = \Delta_{2\text{pt}}^{G^2} + \frac{1}{2} \Delta_{2\text{pt}}^{G^2}, \quad \left| \frac{\delta \Gamma}{\Gamma} \right|_{\text{env}} \simeq 2 \left| \frac{\delta g}{g} \right|_{\text{env}}.$$

Numerically this gives a median conservative envelope $|\delta g/g|_{\text{env}} = 0.058$ and $|\delta \Gamma/\Gamma|_{\text{env}} = 0.115$, with maxima 0.128 and 0.257, respectively, over the grid.

Therefore the screen was not small enough to justify silently dropping G^2 at publication level. It did not prove that the radiative three-point G^2 correction was as large as the envelope; it showed that a few-percent amplitude effect was plausible in the same heavy-heavy OPE window. This diagnostic is superseded by the explicit all-positive G^2 integration below.

The channel-by-channel conservative envelope is summarized in Table 5. The table is generated by `bc_ps_g2_corrected_summary.py` and should be interpreted as a systematic bound, not as the final explicit radiative G^2 contribution.

Table 5: Perturbative $B_{c1} \rightarrow B_c \gamma$ result with conservative G^2 width envelope. Widths are in keV; the last column gives median/max envelope sizes.

Channel	$g_{\text{pert}} [\text{GeV}^{-1}]$	Γ_{pert}	G^2 env.
$B_{c1}(6743) \rightarrow B_c \gamma$	0.0316[0.0269, 0.0370]	0.223[0.162, 0.308]	$\pm 0.025 / \pm 0.060$
$B_{c1}(6750) \rightarrow B_c \gamma$	-0.184[-0.205, -0.167]	8.01[6.60, 9.90]	$\pm 0.914 / \pm 2.30$

Denominator-Reduced G^2 Inventory

To test whether the full explicit radiative G^2 correction could be obtained from ordinary positive-power Schwinger integrals alone, we reduced the projected G^2 numerators in inverse propagator denominators d_1, d_2, d_3 . This is done by `step3_ps_g2_denominator_reduction.wl`; the summary is generated by `step3_ps_g2_denominator_reduction_summary.py`.

The complete reduction contains 747 terms. Of these, 375 have all effective denominator powers positive, while 372 terms have at least one nonpositive effective power and therefore belong to the contact/derivative sector. Thus the latter sector is not a small correction to the bookkeeping; it is roughly half of the denominator-reduced expression.

Consequently, an integration of only the all-positive sector would not be defensible without auditing the contact sector. The refined audit below shows that the contact rows do not carry ordinary two-channel double-Borel support, so the final explicit numerical correction is the all-positive sector reported in the next section.

$B_c^* \gamma$ Denominator-Reduced G^2 Inventory

The same denominator-reduction audit was carried out for the vector final state. The projected structure is

$$V_{\mu\nu\rho} = p_\nu \epsilon_{\mu\rho pq} - (p \cdot q) \epsilon_{\mu\nu\rho p}, \quad V_{\mu\nu\rho} V^{\mu\nu\rho} = -4p^2(p \cdot q)^2.$$

The script `step3_vec_g2_denominator_reduction.wl` replaces the appropriate propagator numerators by $S_Q^{(G^2)}$ or by open one-gluon pieces $S_Q^{(G)} S_{Q'}^{(G)}$, projects onto $V_{\mu\nu\rho}$, and rewrites the result in powers of d_1, d_2, d_3 . The summary script is `step3_vec_g2_denominator_reduction_summary.py`.

The complete vector reduction contains 1095 terms:

sector	number of terms
all-positive denominator powers	538
contact/derivative sector	557

By current, the split is

current	all-positive	contact/derivative
J_μ^A	200	148
J_μ^B	338	409

and by background-field class it is

class	all-positive	contact/derivative
single-line G^2	209	368
open-pair GG	329	189

Thus the contact/derivative sector is slightly larger than the ordinary positive-denominator sector in the vector channel. This is why the $B_c^*\gamma$ result should be quoted as a leading perturbative vector baseline with a G^2 /contact systematic screen, rather than as a completed full-OPE prediction.

The support-level double-Borel audit is performed by `step3_vec_g2_contact_support_audit.py`. It first separates contact rows that contain both hadronic virtualities from rows that are single-virtuality or no-external-virtuality subtraction terms. The first-pass result is

contact support class	number of rows
double-Borel-active candidates	131
single-virtuality rows	309
no-external-virtuality rows	117

All 131 first-pass active candidates come from the anti-bottom photon topology with remaining denominator support $\{d_1, d_3\}$. This needs one more physical-support check. In that topology

$$d_1 = k^2 - m_c^2, \quad d_3 = (k - p + q)^2 - m_b^2.$$

Combining only d_1 and d_3 with Schwinger parameters gives an external piece proportional to

$$\frac{t_1 t_3}{t_1 + t_3} [-P^2 + 2p^2] = \frac{t_1 t_3}{t_1 + t_3} (p - q)^2, \quad P = p + q, \quad q^2 = 0.$$

Thus the two-denominator contact candidate depends on the crossed virtuality $(p - q)^2$, not on independent physical initial- and final-channel pole variables. The refined audit `step3_vec_g2_contact_borel_admissibility.py` gives

refined contact support class	number of rows
crossed single-combination rows	131
single-virtuality rows	309
no-external-virtuality rows	117
ordinary two-channel contact rows	0

Therefore no contact/derivative row is added directly to the physical double-pole numerical sum in the standard two-channel Borel projection. The 131 crossed rows are stored in `step3_vec_g2_contact_active_candidates.csv` as an audit trail; they would require a separate crossed-contact prescription before being used in a numerical width.

For a student checking this calculation, the most useful debugging sequence is:

1. verify the projector normalization $V_{\mu\nu\rho} V^{\mu\nu\rho} = -4p^2(p \cdot q)^2$;
2. reproduce one J_μ^A single-line insertion before running all topologies;
3. check the denominator substitutions separately for charm-photon and anti-bottom-photon emission;

4. classify terms with effective powers $n_i > 0$ as ordinary Schwinger terms and terms with at least one $n_i \leq 0$ as contact/derivative terms;
5. run the support audit, then the refined Borel-admissibility audit;
6. check that the apparent 131 active rows all reduce to crossed $(p - q)^2$ support;
7. keep these rows in the audit trail, but do not add them to the physical double-pole sum unless a separate crossed-contact prescription is introduced.

10 Explicit All-Positive G^2 Integration

The old screening estimate is now superseded by the explicit all-positive dimension-four calculation. After reducing the projected G^2 numerators in inverse denominator powers d_1, d_2, d_3 , the ordinary part is the sector with

$$n_1 > 0, \quad n_2 > 0, \quad n_3 > 0.$$

These rows have standard Schwinger representations and are Borel transformed directly. The pseudoscalar channel contains 375 such terms and the vector channel contains 538 such terms.

The heavy-heavy convention is the same as in the submitted B_c -mixing analysis:

$$S_Q^{(G^2)} : \frac{G^2}{12}, \quad S_Q^{(G)} S_{Q'}^{(G)} : \frac{G^2 N_c^2 - 1}{96 \cdot 2}, \quad N_c = 3.$$

The direct numerical check uses the diagonal double-Borel relation

$$M_1^2 = M_2^2 = 2M^2,$$

fixes $p^2 = m_f^2$, and uses

$$p \cdot q = \frac{m_i^2 - m_f^2}{2}$$

for each physical initial state. The scripts are `bc_ps_g2_all_positive_direct_borel.py`, `bc_ps_g2_all_positive_mc_update.py`, `bc_vec_g2_all_positive_direct_borel.py`, and `bc_vec_g2_all_positive_mc_update.py`.

The pseudoscalar contact audit gives

refined contact support class	number of rows
crossed single-combination rows	80
single-physical-virtuality rows	220
no-external-virtuality rows	72
ordinary two-channel contact rows	0

Together with the vector audit above, this means that no contact/derivative row is added to the standard physical double-Borel sum.

The all-positive G^2 correction shifts the couplings by

channel	$\delta g/g$ from G_{ap}^2
$B_{c1}(6743) \rightarrow B_c \gamma$	+0.543[+0.537, +0.546]
$B_{c1}(6750) \rightarrow B_c \gamma$	+0.120[+0.112, +0.130]
$B_{c1}(6743) \rightarrow B_c^* \gamma$	-0.0245[-0.0328, -0.0190]
$B_{c1}(6750) \rightarrow B_c^* \gamma$	+0.0298[+0.0264, +0.0347]

and gives the final Monte Carlo-folded results

channel	g	Γ (keV)
$B_{c1}(6743) \rightarrow B_c \gamma$	+0.0487[0.0414, 0.0572] GeV^{-1}	0.532[0.385, 0.734]
$B_{c1}(6750) \rightarrow B_c \gamma$	-0.206[-0.230, -0.188] GeV^{-1}	10.04[8.29, 12.45]
$B_{c1}(6743) \rightarrow B_c^* \gamma$	-0.0764[-0.0821, -0.0717] GeV^{-2}	70.3[61.9, 81.2]
$B_{c1}(6750) \rightarrow B_c^* \gamma$	+0.173[0.161, 0.185] GeV^{-2}	382[331, 437]

11 Vector Projector and Width Normalization Audit

The vector tensor used in the projection,

$$V_{\mu\nu\rho} = p_\nu \epsilon_{\mu\rho pq} - (p \cdot q) \epsilon_{\mu\nu\rho p},$$

has mass dimension three. Consequently the coefficient extracted from the sum rule, denoted here by g_V , has dimension GeV^{-2} , not GeV^{-1} . The script `bc_vec_projector_width_audit.py` performs the explicit polarization sum and gives

$$\frac{1}{3} \sum_{\text{pol}} |\eta_A^\mu \epsilon_\gamma^\nu \eta_V^\rho V_{\mu\nu\rho}|^2 = \frac{2}{3} m_i^2 E_\gamma^2 (2m_f^2 + E_\gamma^2).$$

The vector-width formula used in the numerical code is therefore

$$\Gamma(1^+ \rightarrow 1^- \gamma) = \frac{\alpha_{\text{em}}}{3} g_V^2 (2m_f^2 + E_\gamma^2) E_\gamma^3.$$

If one wants the same one-momentum E1 convention as in $\Gamma = \alpha_{\text{em}} g_{E1}^2 E_\gamma^3 / 3$, then

$$g_{E1} = \sqrt{2m_f^2 + E_\gamma^2} g_V.$$

Numerically the conversion factor is 8.969 for both vector channels, giving

channel	g_V (GeV^{-2})	g_{E1} (GeV^{-1})
$B_{c1}(6743) \rightarrow B_c^* \gamma$	$-0.0764[-0.0821, -0.0717]$	$-0.685[-0.736, -0.643]$
$B_{c1}(6750) \rightarrow B_c^* \gamma$	$+0.173[0.161, 0.185]$	$+1.55[1.45, 1.66]$

Thus the projector and width normalization are internally consistent. The large 6750/6743 vector hierarchy is not caused by a width-conversion factor; it comes from the mixed-current vector amplitude and should be checked through the separate sign, current-assignment, and no-mixing-limit audits.

12 Vector Mixing/Sign Audit

The script `bc_vec_mixing_sign_audit.py` tests the remaining convention ambiguity in the $B_c^* \gamma$ channel. At the central point $M^2 = 8 \text{ GeV}^2$, $s_0 = 54 \text{ GeV}^2$, it finds

$$\hat{\Pi}_A = 0.00637, \quad \hat{\Pi}_B \simeq -0.00848, \quad \theta = 43.29^\circ.$$

The baseline mixed amplitudes are therefore

$$\begin{aligned} \hat{\Pi}_{6743}^{(V)} &= \sin \theta \hat{\Pi}_A + \cos \theta \hat{\Pi}_B = 0.00437 - 0.00617 = -0.00180, \\ \hat{\Pi}_{6750}^{(V)} &= \cos \theta \hat{\Pi}_A - \sin \theta \hat{\Pi}_B = 0.00464 + 0.00582 = 0.01046. \end{aligned}$$

The lower vector amplitude is thus cancellation-dominated, while the higher one is constructive.

The nine-point grid medians for several convention tests are

test	Γ_{6743} (keV)	Γ_{6750} (keV)	$\Gamma_{6750}/\Gamma_{6743}$
baseline	70.2	338	4.82
radiative J_B sign flip	2402	4.29	0.0018
alternate rotation (c, s)	29.8	344	11.5
alternate rotation ($c, -s$)	2361	10.1	0.0043
swap eigenstate assignment	316	75.3	0.239
swap assignment and J_B flip	3.99	2572	645

The no-mixing diagnostic is even more transparent. It is not a physical prediction; it only asks what the basis currents would give before forming the mixed states. At the same central point,

diagnostic	Γ_{6743} (keV)	Γ_{6750} (keV)	$\Gamma_{6750}/\Gamma_{6743}$
pure J_A	878	126	0.143
pure J_B	1554	222	0.143
$\theta = 0$: low = J_B , high = J_A	1554	126	0.0808
$\theta = 90^\circ$: low = J_A , high = $-J_B$	878	222	0.253

Thus the no-mixing limits show the same qualitative lower-state-dominant hierarchy seen in representative model calculations. The reversed hierarchy in the baseline mixed-current result is therefore a genuine interference effect: the 6743 amplitude is suppressed by cancellation, while the 6750 amplitude is enhanced constructively.

Representative model calculations have high/low < 1 in the $B_c^*\gamma$ channel. The baseline QCDSR convention instead has high/low > 1 , so we also performed an independent tensor-current sign check. FeynCalc defines

$$\text{DiracSigma}(\gamma_\mu, \gamma_\alpha) = \frac{i}{2}[\gamma_\mu, \gamma_\alpha],$$

so the current factor $i\sigma_{\mu\alpha}P^\alpha\gamma_5$ is equivalent to

$$-\frac{1}{2}[\gamma_\mu, \gamma_\alpha]P^\alpha\gamma_5.$$

The script `step2_vec_tensor_sign_independent_audit.wl` recomputes the projected J_B triangle with both representations. After stripping the common γ_5/ϵ phase, it finds

tensor-current test	ratio to working J_B
working $i\text{DiracSigma}$	+1
explicit commutator $-[\gamma_\mu, \gamma_\alpha]/2$	+1
opposite current phase	-1
reversed sigma indices	-1
no raw i multiplying DiracSigma	$-i$

Thus a radiative J_B sign flip would be a different tensor-current phase convention, not a correction of the present trace. Within the state assignment and current convention inherited from the two-point B_c -mixing sum rules, the vector-channel hierarchy is therefore a mixed-current interference effect.

13 Reproduction Checklist for the Present Pilot

The current numbers can be reproduced in the following order.

1. Run `Bc_gamma/scripts/step1_hard_photon_numerators.wl`. This computes the raw FeynCalc traces for photon emission from the charm line and from the anti-bottom line.
2. Run `Bc_gamma/scripts/step2_ps_e1_triangle_reduction.wl`. This projects the $B_c\gamma$ traces onto $S_{\mu\nu} = p_\nu q_\mu - (p \cdot q)g_{\mu\nu}$ and checks $S_{\mu\nu}\mathcal{P}^{\mu\nu} = 1$.
3. Run `Bc_gamma/scripts/step2_vec_epsilon_A_reduction.wl` and `Bc_gamma/scripts/step2_vec_epsilon_B_reduction.wl`. These produce the diagnostic $B_c^*\gamma$ cores and check the normalization of $V_{\mu\nu\rho}$.

4. Run `Bc_gamma/scripts/step2_vec_tensor_sign_independent_audit.wl`. This verifies that the working `iDiracSigma` tensor current agrees with the explicit commutator form $-\gamma_\mu, \gamma_\alpha P^\alpha \gamma_5/2$.
5. Insert the compact projected cores into `Bc_gamma/scripts/bc_ps_complete_analysis.py`. The script uses the diagonal calibration $M_1^2 = M_2^2 = 2M^2$, with $M^2 = \{7, 8, 9\} \text{ GeV}^2$ and $s_0 = \{53, 54, 55\} \text{ GeV}^2$.
6. Extract f_1 and f_2 from the diagonalized two-point perturbative moments and then sample $m_b, m_c, M^2, s_0, \theta$, and f_{B_c} in the Monte Carlo.
7. Rotate the basis amplitudes with $\hat{\Pi}_{6743} = \sin \theta \hat{\Pi}_A + \cos \theta \hat{\Pi}_B$ and $\hat{\Pi}_{6750} = \cos \theta \hat{\Pi}_A - \sin \theta \hat{\Pi}_B$.
8. Run the all-positive G^2 direct-Borel scripts and fold the median $\delta g/g$ correction into the same Monte Carlo ensemble.
9. Convert the final couplings to widths using

$$\Gamma(1^+ \rightarrow 0^- \gamma) = \frac{\alpha_{\text{em}}}{3} g^2 \left(\frac{m_i^2 - m_{B_c}^2}{2m_i} \right)^3.$$

At this stage both the $B_c \gamma$ and $B_c^* \gamma$ entries include the explicit all-positive G^2 sector. The contact audits show no ordinary two-channel contact contribution; crossed-contact rows are kept separately as a consistency check.

14 Double-Dispersion Denominator Check

The double-dispersion variables satisfy

$$P^2 = p^2 + 2p \cdot q, \quad pq = \frac{P^2 - p^2}{2}.$$

For $B_c \gamma$, the structure is close to the strange-sector E1 case, but with two heavy masses. For $B_c^* \gamma$, the additional $1/p_2$ and pq/p_2 structures make the double-dispersion bookkeeping a useful independent check of the diagonal Borel implementation.

Double-dispersion denominator

For the photon emitted from the charm line, the three denominators are

$$D_1 = (k + q)^2 - m_c^2, \quad D_2 = k^2 - m_c^2, \quad D_3 = (k - p)^2 - m_b^2.$$

With Feynman parameters x, y, z , $x + y + z = 1$, the combined denominator is

$$xD_1 + yD_2 + zD_3 = \ell^2 - \Delta_c,$$

where

$$\Delta_c = (1 - z)m_c^2 + zm_b^2 - zyp^2 - xzP^2.$$

For photon emission from the $\bar{b}$ line, using the momentum routing in the FeynCalc scripts,

$$D_1 = k^2 - m_c^2, \quad D_2 = (k - p)^2 - m_b^2, \quad D_3 = (k - p + q)^2 - m_b^2,$$

and the analogous reduction gives, with $x + y + z = 1$,

$$\Delta_{\bar{b}} = xm_c^2 + (1 - x)m_b^2 - x(1 - x)p^2 + xz(P^2 - p^2).$$

Equivalently one may relabel the Feynman parameters after choosing a different loop momentum. These forms make clear why the true radiative problem is a double-dispersion problem in p^2 and P^2 , not simply the two-point mixing correlator with an extra charge factor.

References

- [1] R. Aaij et al. (LHCb Collaboration), “Observation of Orbitally Excited B_c^+ States,” Phys. Rev. Lett. 135, 231902 (2025).
- [2] R. Aaij et al. (LHCb Collaboration), “Study of $B_c(1P)^+$ states in the $B_c^+\gamma$ mass spectrum,” Phys. Rev. D 112, 112003 (2025).
- [3] B. Martín-González, P. G. Ortega, D. R. Entem, F. Fernández, and J. Segovia, “Towards the discovery of novel B_c states: radiative and hadronic transitions,” arXiv:2205.05950.
- [4] A. E. Bondar and A. I. Milstein, “Phenomenology of D_{s1} mesons radiative transitions,” Phys. Rev. D 111, 114019 (2025).
- [5] S. Godfrey, Phys. Rev. D 70, 054017 (2004).
- [6] D. Ebert, R. N. Faustov, and V. O. Galkin, Phys. Rev. D 67, 014027 (2003).
- [7] L. P. Fulcher, Phys. Rev. D 60, 074006 (1999).
- [8] S. S. Gershtein, V. V. Kiselev, A. K. Likhoded, and A. V. Tkabladze, Phys. Rev. D 51, 3613 (1995).
- [9] T. Y. Li, L. Tang, Z. Y. Fang, C. H. Wang, C. Q. Pang, and X. Liu, Phys. Rev. D 108, 034019 (2023).
- [10] Q. Li, M. S. Liu, L. S. Lu, Q. F. Lü, L. C. Gui, and X. H. Zhong, Phys. Rev. D 99, 096020 (2019).
- [11] X. J. Li, Y. S. Li, F. L. Wang, and X. Liu, Eur. Phys. J. C 83, 1080 (2023).
- [12] E. J. Eichten and C. Quigg, Phys. Rev. D 99, 054025 (2018).